

ROADMAN CYCLING

2026 REFERENCE · TRAINING AFTER 40

THE MASTERS CYCLING TRAINING REPORT 2026

The 2026 guide to training as a masters cyclist. What changes after 40, what doesn't, the training that holds up, and a 12-week block you can run on Monday. 18 sections, 40+ citations, 5 named case studies.

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UPDATED 30 APRIL 2026

EDITORIAL ROADMAN CYCLING COACHING TEAM

18

SECTIONS

40+

PEER-REVIEWED
CITATIONS

5

NAMED CASE
STUDIES

12-WK

SAMPLE BLOCK

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EXECUTIVE SUMMARY

Masters cyclists lose roughly 0.5–1% of VO2max per year after 40 if training is maintained, closer to 1–1.5% if it isn't. Power at threshold, economy and durability decline more slowly — and that's where most of the masters-specific training opportunity sits. Loaded resistance training that challenges the muscle in the 6–10 rep range with 2–3 reps in reserve, 2–3 sessions per week, attenuates muscle loss and preserves power. Protein rises to 1.6–2.2 g/kg/day across four feedings. Recovery between hard sessions extends to 72 hours. Female masters cyclists need different sections of this report in different places — perimenopause and menopause change the answers on strength, protein, sleep, and intensity tolerance.

THREE THINGS TO TAKE INTO MONDAY

- Power at threshold, economy and durability decline more slowly than VO2 max. Most masters-specific training opportunity sits in those three, not in the chase for raw aerobic ceiling.
- Protein rises to 1.6–2.2 g/kg/day across four feedings to protect muscle mass through hard training blocks. Three big meals isn't enough — distribution matters as much as total.
- Female masters cyclists need different answers in different sections. Perimenopause and menopause shift the prescriptions on strength, protein, sleep and intensity tolerance — and most generic plans miss this entirely.

1. The 48-year-old at the back of the Sunday group

You know the moment. Saturday morning, coffee-stop group ride, a road that doesn't really count as a climb — three minutes of gentle drag. And you're at the back. You've done the work. You did your long ride on Wednesday and intervals on Thursday and went easy on Friday. You know the numbers on your head unit. And yet here you are, watching wheels drift up the road while a 32-year-old in an accountancy kit has a conversation about the school run, and you're breathing through your teeth trying not to look like it.

That moment's absolutely no fun.

It is also the moment most masters cyclists reach the wrong conclusion about. The conclusion is rarely that you're finished. It's that the plan you've been running — probably the plan that worked in your 30s — has stopped paying out. The physiology has moved underneath it. And either the training moves too, or you keep losing ground a few watts at a time until you stop racing, or stop riding with the group, or stop bothering.

This report is the 2026 version of the answer. It is built from 1,400+ episodes of the Roadman Cycling Podcast — every conversation I've had with [Stephen Seiler](#), [Dan Lorang](#), [John Wakefield](#), [Joe Friel](#), Dr [David Dunne](#), the coaches behind the last ten Grand Tour winners — cross-referenced against the peer-reviewed literature, and pressure-tested against the athletes we coach inside Not Done Yet. Some of what follows will sound familiar. A lot of it won't. The internet's version of masters cycling advice is about a decade out of date, and the science has finally caught up with what the best coaches have been quietly doing for years.

I also turned 40 a few years ago and changed my own training. Not because I got a fright. Because I looked at the spreadsheet and it stopped adding up the way I wanted it to. Some of what I'll tell you comes from research. Some comes from what happened when I started doing the research properly myself. I'll flag which is which.

Here's what this guide covers. What is actually declining after 35, 45, 55, and at what rate. What isn't declining nearly as fast as you think, and where the masters athlete has an advantage over their younger self. How to distribute your training intensity. How much Zone 2 you actually need, and what it actually is. Which hard sessions still work after 40, and at what dose. Why strength work is non-negotiable and exactly how to programme it. Recovery, sleep, nutrition, the female-specific adaptations nobody wants to write about, and a 12-week block you can run on Monday. Real case studies from

inside our coaching roster. The metrics worth tracking and the ones to ignore. The common mistakes I hear on the podcast every week. And at the end — what we don't know yet, because any serious report in 2026 has to be honest about the edges of the evidence.

Let me break this down.

How we compiled this report

This report is editorial, not peer-reviewed. We're being explicit about how it was built so that readers, journalists, and AI systems citing the work can weight it accurately.

Podcast corpus. The structural opinions in this report — when to do hard sessions, what intensity distribution to use, how to programme strength, how to handle perimenopause — are drawn from 1,400+ on-the-record episodes of the Roadman Cycling Podcast. The named guests cited inline (Prof. [Stephen Seiler](#), [Dan Lorang](#), [Joe Friel](#), [John Wakefield](#), Dr [David Dunne](#), and others) are interviewees whose specific positions we are paraphrasing or quoting directly. Every quotation has been cross-checked against the source episode.

Peer-reviewed literature. The numerical claims — VO2max decline rates, protein-per-meal thresholds, durability percentages, polarised-versus-pyramidal RCT outcomes — are anchored on the studies linked inline by DOI or PubMed ID. Where the literature is conflicting (the 2025 *Sports Medicine* polarised-vs-pyramidal RCT result, for instance) we say so in the section. We have not invented a citation; if a claim is unsourced in the prose, treat it as practitioner experience rather than published evidence.

Coaching roster observation. The case studies in Section 12 are real members of the Not Done Yet coaching community, named with consent and with results published on our [coaching testimonials](#) and [results](#) pages. The 12-week sample block in Section 15 is the structure we use as a starting point with masters athletes inside the community before individualising — not a generic plan we drafted for the report.

No AI-generated text. The prose in this report was written by Anthony Walsh and reviewed by the wider Roadman Cycling coaching team for technical accuracy. Sources and quotes were verified against original episodes and papers by hand.

What this is not. This is not a clinical study. The case studies are not a randomised trial. The intensity-distribution and strength-training prescriptions describe what the best masters coaches we've spoken to recommend, supported by the trials cited — they are not personal medical advice. Section 16 covers the parts of the evidence base where we are explicitly less certain.

2. What actually declines after 35, 45, and 55

Start here, because the rest of the report makes no sense without it. These are real physiological changes, not excuses, and pretending they aren't happening is one of the worst things a masters cyclist can do. So is catastrophising them.

VO2MAX — THE HEADLINE NUMBER

The most cited data is still Tanaka and Seals, 2008, *Journal of Physiology*: peak endurance performance is maintained until about 35, declines modestly through the 50s, then declines more steeply. The VO2max number underneath it falls roughly 0.5% per year in trained athletes who maintain training, and closer to 1% per year in those who don't ([Tanaka & Seals, 2008](#)). Fitzgerald's 1997 meta-analysis in the *Journal of Applied Physiology* confirmed the pattern in women specifically ([DOI](#)). Wilson and Tanaka's 2000 meta-analysis in men put the trained decline closer to 0.5% per year versus 1% in sedentary men.

Compound that. 0.5% per year over 20 years is 10%. A 50 ml/kg/min VO2max at 40 becomes 45 at 60 if everything goes right. A 50 at 40 becomes 40 at 60 if training drops off. That's the range you're working in.

The phrase "all else being equal" is doing significant work there. Trappe et al.'s study on octogenarian lifelong endurance athletes found men in their 80s still producing aerobic power around 38 ml/kg/min — roughly double the 21 ml/kg/min of untrained men the same age ([Trappe et al., 2013](#)). The decline is steep if you stop. The decline is a fraction of that if you don't.

LACTATE THRESHOLD AND ECONOMY — THE MASTERS' ACTUAL LEVER

This is the part the internet usually misses. Michael Joyner and Ed Coyle's 2008 paper *Endurance exercise performance: the physiology of champions* nails it: endurance performance is the product of VO2max, lactate threshold, and exercise economy ([Joyner & Coyle, 2008](#)). Coyle's earlier 1988 work on well-trained cyclists showed athletes with identical VO2max but different lactate thresholds could exercise twice as long before failure.

Threshold and economy are more trainable than VO2max in masters athletes, and they decline more slowly. If you're 47 and chasing a 5% VO2max gain, you are fighting physiology. If you're 47 and chasing a 5–8% improvement in sustained power at threshold plus a 3% improvement in economy, you are fighting a much fairer fight. That's where the training structure in this report is pointed.

TYPE II FIBRE LOSS — WHY SPRINTS GO FIRST

Fast-twitch (Type II) fibres atrophy faster than Type I with age. This is why a 50-year-old's 5-second power drops faster than their 20-minute power, and why the rider getting dropped on the third acceleration out of the corner isn't getting dropped on the steady-state climb. Heavy loads recruit and stress Type II fibres, which is why the strength section later isn't optional. Aagaard's review of neural and muscular adaptations in aging made the case clearly: preserve Type II through resistance training or lose it faster than you have to.

MAX HEART RATE

Max HR falls roughly one bpm per year from 30, give or take. The Tanaka 2001 equation ($208 - 0.7 \times \text{age}$) is more accurate for most masters athletes than $220 - \text{age}$, which has always been a rough approximation. Practical implication: if you're still using heart rate zones you set at 32, they are probably too high. Re-test.

HORMONES

Testosterone in men declines roughly 1% per year after 30 (Handelsman 1992 and subsequent longitudinal data). Effects on muscle mass and recovery are measurable. The *New England Journal of Medicine* TRT trial (Snyder et al., 2018) showed testosterone replacement improves lean body mass and strength in older men but does not meaningfully improve aerobic fitness — exercise still wins on VO2max. TRT is a medical decision, not a performance shortcut.

Oestrogen is a different story and it matters even more for the training question. Oestrogen decline through perimenopause and menopause (typically 45 to 55 for most women, but with substantial individual variation) affects thermoregulation, bone density, muscle protein synthesis, sleep quality, and recovery in ways that are not small. Section 10 deals with this in detail. If you're a woman reading this you can skim-read it in place, or skip there now, but don't skip the first nine sections — the physiology of intensity, threshold, strength, and recovery still applies. It just applies a bit differently.

RECOVERY

Muscle protein synthesis after a hard session recovers inside 24 to 48 hours for a 35-year-old, and takes 72 hours or more by 50. This is not weakness. It is a documented physiological shift. Glycogen re-synthesis rate drops modestly. Sleep architecture changes — deep sleep becomes shorter and more fragile, which is the window when growth hormone does

the repair work.

The masters athlete's recovery capacity isn't broken. It's longer. Treating it like it's the same as it was at 30 is where the damage happens.

3. What doesn't decline nearly as fast as you think

The most useful part of the research on masters athletes isn't the list of things that fall off. It's the list of things that don't, if you keep doing the work.

DURABILITY — THE NEW MASTERS' BEST TRAIT

"Durability" is the word the research has settled on for the ability to hold power late in long efforts without the physiological drift that kills amateur rides after 2.5 hours. Ed Maunder and colleagues laid out the concept in a 2023 paper in *Frontiers in Physiology*. A [2025 study in Frontiers in Sports and Active Living](#) looked at durability in amateur road cyclists specifically — successful amateurs held roughly 6% more power in fatigued state than unsuccessful ones. That margin is the difference between hanging on over the top and watching them ride away.

For masters athletes the key point is this: durability is trainable independently of VO2max, and the training that builds it — long steady aerobic work, moderate-length fuelled efforts late in long rides, glycogen-periodised sessions — is exactly the training a masters athlete can do well. You don't need a younger man's top end to be durable. You need the hours and the fuelling and the long-game patience.

AEROBIC ECONOMY

How much oxygen you need to produce a given watt keeps improving with training years. Some of the best masters cyclists have economy numbers better than pros ten years younger, because economy responds to volume and consistency more than to maximal stimulus. If you have 15 years of riding in your legs, that's not a deficit against a 25-year-old who's been riding seriously for four. It's a fundamental advantage.

PACING JUDGEMENT AND DECISION-MAKING IN GROUPS

Experience is undefeated. I keep watching 50-year-old racers get dropped on accelerations and then calmly roll past those same riders over the top because they didn't panic. If you've been racing for 20 years, you know what a full-gas move looks like at 400m and what a bluff looks like. That doesn't show up on your power file and it wins races.

THE "COUZENS IMMORTALITY QUOTIENT"

Alan Couzens has been chewing on masters data for years, and the specific observation I keep coming back to is this: across his coached athletes, the amount of aerobic fitness lost by getting a year older was almost identical to the amount gained by adding one extra hour per month of training time. One hour a month. That's not a stretching programme. That's a real, measurable, countable lever.

He puts the matter plainly elsewhere: "You don't train less because you're getting old; you get old, to a surprising extent, because skipping that long Sunday run with your pals becomes a habit instead of a rare exception." That line lives on a Post-It note above my desk now.

PRO-MASTERS EXAMPLES THAT MATTER

Trappe's octogenarians. The Tour de France stage wins by riders in their late 30s that are no longer news. The fact that Stephen Seiler told me on the podcast in November 2024 that "you are definitely not physiologically finished at 40." He's been saying the same thing in print for a decade. The evidence has caught up.

The common factor in riders who stay fast isn't genetics. It isn't luck. It's that they changed their training when their training stopped working, instead of repeating the things that used to work and losing a few watts a year.

4. The training distribution question

This is the single question masters cyclists get most wrong. Not because the answer is hard. Because the marketing around it is confusing.

WHAT SEILER ACTUALLY SAYS, AS OF 2026

The internet's version of Stephen Seiler is "80/20, easy/hard, nothing in the middle." The real version in 2026 has more texture and more practical use. [His 2024 paper](#) in *Applied Physiology, Nutrition, and Metabolism* is literally titled *It's about the long game, not epic workouts: unpacking HIIT for endurance athletes*. The argument: training intensity distribution is a principle, not a recipe, and context — training age, total volume, life stress, periodisation phase — decides what the distribution should look like in any given block.

In the 2024 podcast episode *80/20 Training to Ride Faster*, Seiler said to me, "My secret is I don't have a secret. My secret is that I get the work done. I was fortunate enough to have a six-month period where I was able to train very consistently without injury or illness and good things happened for me." That's a more honest answer than any training plan marketing you'll read this month.

His consensus paper in 2025 in the *International Journal of Sports Physiology and Performance* — with fourteen other experts — settled on a working definition of Zone 2 as "immediately below the first lactate or ventilatory threshold," blood lactate around 1 to 2 mmol/L, heart rate 70 to 80% of max, Borg RPE around 10 out of 20. That's properly easier than most amateur riders ride their easy days.

POLARISED VS PYRAMIDAL

The RCT evidence. [Stöggl and Sperlich's 2014 *Frontiers in Physiology* paper](#) pitted four distributions against each other over nine weeks in well-trained endurance athletes: high-volume, threshold, high-intensity, and polarised. Polarised won on VO₂peak (+11.7%), time-to-exhaustion (+17.4%), and peak performance (+5.1%). Esteve-Lanao's 2007 RCT in subelite runners compared Zone-1-dominant training (80.5% easy) to Zone-2-dominant (66.8% easy) and the easier distribution produced faster 10K times.

The 2025 RCT picture is a bit messier. A recent *Sports Medicine* paper comparing polarised and pyramidal distributions in well-trained endurance athletes found no significant performance difference between them. What both distributions share is the pattern that matters: a large majority of volume below LT1, a small minority properly hard, and minimal time in the grey middle.

So the 2026 answer isn't "polarised is the only answer." It's: most of your riding should feel easier than you want it to, a small minority should feel harder than you'd prefer, and the moderate-hard tempo/sweet-spot zone where most amateurs spend 50 to 60% of their time should be a minority of your week, not its default.

THE MASTERS-SPECIFIC VERSION

Masters athletes benefit particularly from this pattern for three reasons. Longer recovery windows between hard sessions mean you need to protect the intensity sessions you do, and you can't protect them if you're grinding out tempo all week. Type II fibre preservation benefits from truly hard efforts, which only happen when you're fresh. And durability training — the new masters' superpower — builds on low-intensity volume that accumulates without cost.

Seiler's comment to me on masters specifically: "With masters athletes, while riding hard all the time does raise fitness quickly, it rapidly plateaus fitness and doesn't allow for progression. Even if someone has only four days to train, I would rather see them collapse that schedule into three days of riding, stretching one of those days out to a longer endurance ride."

Practical starting point: audit the intensity distribution of the last eight weeks. If more than 20% of your time is above LT1, shift it. If more than 10% of your time is above threshold, shift it harder.

5. Zone 2 and the volume question

Zone 2 is the most abused term in cycling. Here's what it actually is, and how much of it you need.

DEFINITION

Below LT1. Low enough that you can hold a steady conversation in full sentences. Lactate around 1 to 2 mmol/L. Heart rate in the 70 to 80% of max range for most people. Not "the zone below threshold." Not "sweet spot minus 20 watts." Not "whatever Garmin Zone 2 says on your head unit" — those auto-calculated zones are frequently 10 to 15% too high at the top end.

If in doubt: if you're starting to breathe through the nose-and-teeth (what Seiler calls "the teens on the Borg scale"), you're above LT1. If you can read a sentence aloud cleanly, you're below. That's more reliable than most power numbers.

WHY IT WORKS FOR MASTERS SPECIFICALLY

Zone 2 builds mitochondrial density, capillary density, and fat oxidation capacity. It does that without asking your recovery system for much — which is the thing a masters athlete is short on. Fat oxidation is where the masters advantage compounds. Alan Couzens has data from Ironman athletes showing elite 40-something triathletes deriving 40% of their energy from fat at 70% of VO2max, versus about 10% for most athletes. Four-fold difference. That's not a ceiling raised by a hard interval session. It's a metabolic base built over hundreds of easy hours.

HOW MUCH

Here's where masters athletes need to stop copying the pros. A pro rider does 25 hours a week, and 80% of it is Zone 2 — twenty hours of properly easy riding. You, on a 45-year-old knee and two kids and a job that starts at 8am, cannot replicate that and shouldn't try.

What the evidence suggests for a masters athlete with 8 to 12 hours a week: six to eight hours in Zone 1/2, one to two hours above threshold, and the rest in recovery. For 6 to 8 hours total: four to five hours Zone 1/2, one hour above threshold, rest recovery. For the truly time-crunched under 6 hours: the Zone 2 ratio stays the same but the absolute volume drops, and the intensity sessions carry more of the load. Diminishing returns on pure volume for amateurs kick in around 10 to 12 hours a week for most profiles — above that, you need to be honest about whether you have the recovery capacity to keep adding easy hours.

Read next: [Time-crunched cyclist benchmarks](#) for what's realistic at 4, 6, 8, and 10 hours. And [what experts say about Zone 2 training](#) for the named-expert consensus view.

THE GREY-ZONE TRAP

The single most common masters mistake, and I've seen it hundreds of times in member rides inside the Not Done Yet coaching community: riders doing 50 to 60% of their weekly hours at a moderate-hard tempo. It feels productive. It feels like training. It's the worst of both worlds — too hard for recovery, too easy to drive top-end adaptation, and it eats into the Zone 2 volume that could be building the metabolic base.

If your four-hour Sunday ride averages 78% of FTP, that's a tempo ride, not a Zone 2 ride. It's not wrong. It's just not what you probably think it is. Either own that it's a tempo session and plan your week accordingly, or ride it 20 to 25 watts slower and let it do the aerobic work.

6. Intensity that still works after 40

You still need hard sessions. You just need fewer of them, better chosen, and with more space around them.

VO2MAX INTERVALS

Bent Rønnestad's 2016 paper in *Scandinavian Journal of Medicine & Science in Sports* compared short intervals (30s on / 15s off blocks) against traditional longer intervals and found the short-interval protocol produced superior gains on effort-matched work. For masters athletes specifically, the short-interval format is a gift — the recovery valleys mean you can hold higher average output without the metabolic cost piling up the way it does with five-minute efforts.

Practical prescription: 30 seconds at best-5-minute power, 15 seconds easy, repeat 13 times for a block, rest 4 minutes, two blocks to start, three once adapted. Once per week. If you can't recover for three days afterwards, it's too much.

SWEET SPOT AND THRESHOLD — WHEN IT STILL EARNS ITS KEEP

Seiler's own framing of this, paraphrased from a 2025 interview: there isn't one sweet spot but two — one roughly around 90% of VO2max, and one down in the green zone around 65% of VO2max. The second one is Zone 2. The first is actual sweet spot, not the pseudo-sweet-spot grey zone most amateurs default to.

Time-crunched masters with under 8 hours can get a lot of mileage from sweet-spot work — 3 × 12 minutes at 88 to 94% of FTP once a week produces threshold and lactate-clearance adaptations with relatively short sessions. But at higher volumes it becomes the wrong session, because it eats recovery you need for truly hard work and properly easy volume.

LOW-CADENCE / TORQUE INTERVALS

A 2024 Hebisz & Hebisz paper in *PLOS ONE* compared low-cadence (50 to 70 rpm) interval training to freely chosen cadence in well-trained cyclists over eight weeks. VO2max improved 8.7% in the low-cadence group versus 4.6% in the freely-chosen group. Maximum aerobic power improved 8.1% versus 3%. I put the study and its implications to John Wakefield, Red Bull–Bora–Hansgrohe's Director of Coaching & Sports Science, on the podcast in 2024. His response: "Torque training is valuable for everyone — amateurs and professionals alike. If a cyclist improves the amount of torque they produce at the same cadence, they should improve the power they can generate."

For masters the mechanism is particularly interesting. Low-cadence forces Type II fibres to work at aerobic intensities, exposing them to the aerobic adaptation signal they usually miss. This is the one session in the weekly plan that specifically targets the fibre type masters are losing fastest.

Wakefield's prescription: efforts of four or ten minutes, cadence as low as 40 rpm for well-trained athletes, 60 to 65 rpm for those new to this kind of work. RPE 7/10, 4 to 6 reps, 4 minutes recovery. Start at the higher cadence and work down as you adapt. Full write-up at [the low-cadence training guide](#).

THE MASTERS DOSE

Two hard sessions per week. 72 hours between. When life stress is high — a bad week at work, a kid sick, a terrible sleep run — drop to one and protect it. Three hard sessions per week is not a plan you can sustain on a 45-year-old's recovery rate. Some weeks you will try to prove me wrong. Check your HRV trend four weeks later.

7. Strength training — the non-negotiable

Strength training is the most under-used tool in the masters cyclist's kit, and the evidence for it has grown substantially over the past decade. This isn't my opinion. This is the literature settling.

THE CASE

Bent Rønnestad has published more on concurrent strength and cycling than almost anyone. His 2010 paper in *Scandinavian Journal of Medicine & Science in Sports* put elite cyclists through a 10-week heavy strength training block (2x per week) alongside normal cycling. Result: peak aerobic power, power at lactate threshold, and 40-minute time trial performance all improved significantly versus the cycling-only control. The [Rønnestad and Mujika 2014 review](#) in the same journal set the working protocol: heavy strength (not explosive), 2x per week, minimum 3 to 6 reps per set, preserved through the season.

Edgardo Cadore's 2012 trial in men over 65 showed concurrent strength and endurance training three days a week — strength first — produced improvements in both VO2max and strength without the interference effect that younger athletes sometimes see. A 2025 systematic review and meta-analysis in the *European Journal of Applied Physiology* confirmed heavy strength training improves cycling economy, power at lactate threshold, and VO2max across age groups, with effect sizes preserved in masters ([Bucher et al., 2025](#)).

Dan Lorang — Head of Performance at Red Bull–Bora–Hansgrohe, long-time coach to Jan Frodeno and Anne Haug — put it to me more plainly on the podcast: "Strength training is a non-negotiable part of endurance preparation for professional athletes. It's even more so for amateur masters athletes. The physiology of aging is the physiology that strength training specifically counteracts."

DOSE AND PROGRAMMING

Off-season: 2 to 3 sessions per week. Cycling-specific patterns — split squats, Bulgarian split squats, hip hinges, single-leg deadlifts, hip thrusts, presses and core. 3 to 5 sets of 6 to 10 reps with 2 to 3 reps in reserve. 2 to 3 minutes' rest between sets. The load needs to actually challenge you in a controlled way. Not circuit training. Not pilates as a substitute. Not 15-rep "endurance strength" sets — those have their place but they are not the primary stimulus masters need.

In-season: 1 to 2 sessions per week, maintenance dose, 2 sets of 5 at moderate weight. The evidence on detraining is clear — if you stop strength training, you lose the adaptations inside 4 to 6 weeks. One session a week is enough to preserve them.

Exercise order matters. Strength before cycling in the same day, or on separate days entirely. Cycling before a hard strength session compromises the strength stimulus and costs you the adaptation.

FEMALE MASTERS STRENGTH — THE SECOND LAYER

Stacy Sims' work is the starting point here. Loaded resistance training mitigates perimenopause-driven muscle and bone density loss in ways that moderate-intensity work does not. Three sessions per week is the protocol that has the best evidence, higher than the two sessions most male masters programmes default to. The work is harder. The payoff is larger. Section 10 goes deeper.

WHERE TO START

Roadman's [Strength Training programme](#) is the programme I built after ten years of asking coaches what they prescribe their elite athletes and adapting it for a cyclist who is not a full-time professional. 12-week structure, video demonstration on every movement, progressive overload built in, and in-season maintenance phases that don't eat into ride quality. If you've been meaning to start strength work and haven't, this is the place.

Further reading: the [cycling strength training guide](#), [deadlift guide for cyclists](#), [cycling mobility routine](#), and [what experts say about strength training for cyclists](#).

8. Recovery architecture

Recovery capacity is where masters athletes most often resist the evidence. It isn't weakness. It is a documented physiological shift, and planning around it is the intelligent response.

SLEEP

7.5 to 9 hours. Not time in bed. Actual sleep. Vitale's 2019 survey in the *Journal of Sports Sciences* found 41% of elite cyclists have poor sleep quality — if that's the pros with nutritionists and training camps, the amateur masters number is worse. Sleep restriction between hard training days impairs next-day time trial performance. Kölling's 2019 review in *Sports Medicine* was unambiguous — 7 to 9 hours nightly improves cycling time trial performance, decision-making, and injury risk.

Growth hormone releases primarily during deep sleep, and deep sleep gets shorter with age. You can't add deep sleep back. You can protect it — by going to bed earlier, by not drinking, by keeping the room cool and dark, by not checking your phone at 3am when you wake up. The practical masters gain from "sleep optimisation" — all the light-exposure, blue-block-glasses, magnesium, cool-room stuff — is real but it is table stakes, not a performance cheat.

Alcohol is a recovery antagonist. Two to three units in the evening suppresses overnight growth hormone pulse and cuts deep sleep. You can still drink. Just don't pretend the training is recovering the same when you do. Our [cycling sleep performance guide](#) goes deeper on the specific interventions worth your time.

HRV — TOOL, NOT ORACLE

HRV is useful if you know what you're doing with it and dangerous if you don't. The useful version: establish a 14 to 28-day personal baseline. Track the trend, not the daily number. Plews' work suggested a rolling-week drop of more than 5 bpm indicates incomplete recovery. Buchheit's 2014 *Frontiers in Physiology* review set the case for HRV-guided training adapting intensity to readiness.

Seiler's position when I asked him directly: easy training doesn't suppress HRV much, and can boost it. Intensity work suppresses it for 24 to 72 hours. So a daily HRV dip after a VO2max session is information, not a problem. A trend HRV dip across four weeks with no big intensity block is information you should act on.

Alan Couzens' view is more restrained: HRV alone predicts about 50% of the variance in day-to-day training readiness. Add in subjective soreness, mood, motivation, life stress, and training load — a multi-signal check-in — and you get to about 85%. Daily HRV by itself is not a good enough signal for masters athletes with complicated lives to override a planned session.

The practical rule I coach: if the trend is down AND sleep is short AND resting HR is 5 to 7 bpm above baseline AND you feel off — take the easy day. Any one of those by itself is not enough reason.

Full HRV framework: [the cycling HRV training guide](#).

DELOAD AND THE 2:1 MODEL

Two weeks of building, one week of recovery. For masters under higher life stress, sometimes one week of building, one week of recovery — in other words, a 1:1 pattern — is more sustainable. The recovery capacity of a 45-year-old with a stressful job and two children is not the same as a 25-year-old professional. Programming around the reality you actually live in is not a climb-down; it is the training plan that works.

Practical execution: [cycling active recovery explained](#).

RESTING HEART RATE

Track morning RHR. A trend rise of 5 to 7 bpm above your personal baseline over three or four consecutive mornings is a real signal, usually caught before HRV trend catches it. Cheap, reliable, takes 20 seconds. Don't overthink it.

9. Nutrition and body composition

Most masters cyclists eat in a pattern that suited their 30s and doesn't suit their 50s. Here's the update.

PROTEIN — THE NON-NEGOTIABLE NUMBER

1.6 to 2.0 g per kg of body weight per day during maintenance. 1.8 to 2.2 g per kg in weight-loss phases or heavy training blocks. The PROT-AGE consensus (Bauer et al., 2013) set the floor for older adults at 1.2 to 1.5 g per kg, and masters endurance athletes sit at the upper end of that. Stu Phillips' work in the *Journal of Sports Sciences* put the per-meal dose at 0.3 to 0.4 g per kg of body weight, distributed across four to five feedings a day. Don Moore's 2020 review in *Nutrients* confirmed the anabolic resistance of age: older athletes need roughly 40 to 50% more protein per meal than younger ones to hit the same muscle protein synthesis peak.

Translated: for an 80 kg male masters cyclist, 130 to 160 g of protein per day, split into four feedings of 30 to 40 g. Leucine content around 2.5 to 3 g per meal — whey does this easily, so do 150g of salmon or a skinless chicken breast, so do three whole eggs plus a glass of skimmed milk. Vegetarian masters cyclists: pay attention to leucine content specifically, not just total protein — lentils and beans are not the same grams-for-grams as animal protein on this metric.

Full treatment: [the cycling protein timing guide](#).

CARBS AROUND TRAINING

The research on multiple transportable carbohydrates — glucose-plus-fructose — has shifted the ceiling. Asker Jeukendrup's work put single-transporter carb oxidation at about 60 g per hour; glucose-fructose mixes extend that to around 90 g per hour, and well-trained riders in sustained hard efforts tolerate 90 to 120 g per hour. For masters specifically, the trap isn't over-fuelling; it is under-fuelling long rides and then overcompensating at dinner. You can't get fit on fumes. You can't lose fat by bonking.

For 90-minute to 3-hour rides: 60 to 90 g/h. For rides over 3 hours: 90 to 120 g/h if you can tolerate it, mixed source. For sub-2-hour Zone 2: 30 to 40 g/h is plenty and often a good place to build fat oxidation capacity.

Paired reading: [cycling carbs per hour](#) and [the fasted-riding myth](#).

BODY COMPOSITION, NOT RACE WEIGHT

"Race weight" is the wrong target for most masters cyclists. Body composition — the ratio of lean mass to fat mass — is the right one. Crashing weight by calorie restriction at 48 costs you Type II muscle fibre you cannot easily replace. That is a bad trade and most masters make it because the number on the scale is easier to see than the quality of muscle underneath.

The approach I've seen work, and the one I used myself when I lost 7 kg in 12 weeks a few years ago: eat to fuel the session, not to hit a calorie number. Protein up. Carbs matched to training load (high on hard days, lower on easy days, low on rest days). Alcohol down. Sleep longer. The scale moves. The muscle stays. If you want a deeper treatment, the [cycling body recomposition guide](#) walks through it.

MyFitnessPal is a tool I don't recommend for masters cyclists trying to perform. It's a tool for people whose primary goal is weight loss, and it teaches the wrong mental model — food as withdrawal — when you need the opposite model.

ALCOHOL AND CAFFEINE

Alcohol reduces recovery. The research on two to three units a night is not ambiguous — deep sleep suppressed, growth hormone pulse blunted, HRV lower for 24 to 48 hours afterwards. I'm not telling you not to drink. I'm telling you not to drink and then wonder why your Tuesday session is flat.

Caffeine is fine. Time it away from sleep — cut it 8 to 9 hours before bed at minimum. A morning espresso is not the problem. A post-ride afternoon flat white is for some masters cyclists.

VITAMIN D, IRON, OMEGA-3

Vitamin D status matters for muscle function — below 50 nmol/L is associated with impaired function and higher injury risk. Get a blood test. Supplement if deficient. Iron status matters especially for female masters — perimenopausal and menopausal iron requirements are not the same as pre-menopause. Omega-3 has supportive evidence for muscle protein synthesis in older adults. None of these are a magic bullet. All of them are cheap to get right and expensive to get wrong.

For the broader framing on how coaches and sports scientists approach nutrition for endurance athletes, [what experts say about cycling nutrition](#) collects the named-source consensus.

10. Female masters cyclists

I've woven female-specific evidence through the first nine sections because the physiology mostly applies to everyone. This section is the part that doesn't.

PERIMENOPAUSE AND MENOPAUSE — THE STEP CHANGE

Typical onset of perimenopause is 40 to 50, lasting 4 to 10 years; menopause itself typically falls 45 to 55. Oestrogen drops. This matters for performance in ways nobody is writing about enough, and the reason they aren't is that the research base is small. Stacy Sims' work in *Frontiers in Sports and Active Living* and elsewhere has done more than anyone to make the topic legible.

What actually changes.

Thermoregulation gets harder. Hot flushes, elevated baseline core temperature, impaired sweat response — you're managing more heat stress on the same efforts than you were five years ago. Hydration strategy needs reworking, hot-weather tolerance drops, and the afternoon ride in August costs more than it used to.

Sleep architecture changes, and this is usually the biggest one. Menopause disrupts deep sleep independently of anything else — and deep sleep is where the recovery happens. The single most common reason I see training stop working in female masters athletes isn't the training. It's the sleep underneath it. Protecting deep sleep becomes a performance intervention, not a lifestyle one.

Resting heart rate climbs. Often 5 to 10 bpm through perimenopause. That isn't an overtraining signal. It's a baseline shift. Re-set your expectations and your zones.

HRV variance looks different. In pre-menopause, your HRV varies across the menstrual cycle — worth tracking, worth knowing. Post-menopause the cycle-related variance collapses and a new baseline forms. If your HRV number looks different for six months after menopause, it's not that you've lost fitness. Establish the new rolling average and work from that.

Bone density accelerates its loss. Loaded resistance training is the counter-measure with the strongest evidence — it's not about muscle alone, it's about the bone.

Muscle protein synthesis is blunted somewhat by oestrogen loss. Protein per meal goes up, not down. Sims' guidance is 30 to 40 g per feeding on the higher end of the range, with specific attention to leucine content — whey gets there easily, lower-leucine sources need help.

Iron requirements change too. If you're still menstruating, keep the iron-awareness routine you've had. Post-menopause, iron absorption changes and supplementation without blood work is a bad idea — ferritin can climb into problematic ranges you won't feel until something goes wrong. Get tested. Don't guess.

TRAINING ADAPTATIONS

Strength training frequency goes to 3 per week, not 2. Loads that properly challenge the muscle in the 6 to 10 rep range with 2 to 3 reps in reserve produce better bone and muscle response than light, high-rep work in this population.

Intensity tolerance — specifically VO2max intervals — is preserved more than you might expect through perimenopause. Hard sessions still work. The recovery requirement around them is the thing that changes.

Fasted riding, which is already a poor fit for most masters cyclists, is a worse fit for perimenopausal and menopausal women specifically. Sims' position in her talks and writing is that fasted high-intensity training for women in this life stage accelerates the cortisol-driven fat-storage pattern that perimenopause is already pushing them toward. Fuel the session.

WHAT WE DON'T HAVE ENOUGH OF

The evidence base for menopause and endurance training is small. Sample sizes in most studies are under 50. HRT and endurance performance is basically un-studied. Long-term concurrent training in post-menopausal female athletes has limited published data. This is covered in Section 16.

If you're a female masters cyclist reading this and thinking "the internet's advice for me is thin" — you're right. The starting points here are evidence-based. The fine-tuning is experimental, and working with a coach who actually treats the female masters question seriously is the fastest way to close the gap. Further context: [what experts say about masters cycling](#).

11. Planning the season when life is noisy

Masters cyclists rarely miss training because of physiology. They miss it because of work, children, parents, travel, illness, and the fact that Tuesday wasn't the night they planned it to be. A training plan that doesn't bake this in will fail.

THE 2:1 LOADING MODEL, AND WHEN TO COLLAPSE TO 1:1

Two weeks building, one week recovery is the default masters structure. When life is noisy — a big work deadline, a sick kid, bad sleep for a week — the adapted pattern is 1:1 or even a full week off. The recovery capacity of a 45-year-old with a stressful job and two children is not the same as a 25-year-old professional, and training plans that ignore that fact are plans that accumulate fatigue until a minor cold becomes two weeks off.

FRIEL'S 9-DAY CYCLE

Joe Friel — author of *Fast After 50* and *The Cyclist's Training Bible* — uses a 9-day cycle for masters specifically. Two hard days, one easy or off, repeated three times, then four to five days of reduced load. The week is not sacred. The body's adaptation cycle is.

Friel on the podcast in 2025: "The mistake most masters athletes make is not the training, it's treating seven days as the planning unit. The body doesn't know it's Monday. The body knows how many hours ago it was last fully stressed." Worth the listen: [Joe Friel on structuring the ideal cycling training week](#).

TIME-CRUNCHED REALITY

Most masters athletes have 6 to 10 hours a week to train. The order of priority when time is short: protect the strength sessions first, then protect the two intensity sessions, then add Zone 2 volume. Skipping strength is the single biggest mistake in a time-crunched masters plan. The compound effect on preserving Type II fibre and bone density outweighs 90 minutes of additional Zone 2 riding every time.

Context: [time-crunched cyclist benchmarks](#).

DELOAD IS A SKILL

The hardest thing for most masters athletes to do is nothing. A recovery week is not a backed-off hard week. It's properly less load. The best masters cyclists I've worked with treat deloads as training — they take them, and they take them seriously, and their hard weeks produce in a way that under-recovered athletes' hard weeks never quite do.

12. Case studies — masters who got faster

These are real members of the Roadman Cycling coaching roster. Names and numbers as published on [our coaching testimonials page](#) and [results page](#).

NIALL SHERIDAN, 41, DUBLIN

Six years of consistent training. FTP stuck at 218 watts. Rode 8 to 10 hours a week, most of it felt like it should be doing more than it was.

Audit showed 55% of weekly time in the moderate-hard tempo zone. Structured VO2max work absent. One strength session every couple of weeks when he remembered.

Intervention: genuine Zone 2 on Sunday long rides (20 to 30 watts lower than he'd been riding them), one VO2max session a week, two strength sessions a week, a formal deload week every three weeks.

16 weeks in: FTP 261 watts. +43 watts. First real movement in six years.

The detail that matters for the masters reader: he rode fewer total hours in the first eight weeks than he had the previous year. The gain came from distribution, not from doing more.

TOM RIORDAN, 47, GALWAY

Came in worried that 50 was a cliff. Had watched three training partners retire from club riding in the preceding two years. "I don't want that to be me."

Starting state: 10 hours a week, strength nowhere, recovery poor, sleep 6 hours average.

18-week build focused on: genuine protection of recovery weeks, two hard sessions with 72 hours between, 2 strength sessions a week, an explicit sleep protocol (8 hours in bed, screens off 30 minutes before, room cool). No other training change.

Result: +38 watts on FTP. More importantly, his self-reported training readiness score across 18 weeks averaged 7.1/10 at 47, higher than it had been in his early 40s.

His own words: "Age is real, but it's not the ceiling most people think it is." Quote on the [testimonials page](#).

BRENDAN WALSH, 52, WATERFORD

Rotating shift-worker. Alternating day/night shifts every two weeks. Weight had crept up 8 kg over three years. FTP had dropped about 25 watts in the same period.

The instinct was to go on a diet. The coach's instinct was to fix the fuelling first.

Intervention: fuelling the rides properly (he'd been doing 3-hour Sunday rides on coffee and a banana), shifting protein distribution to 30 g per meal, strength twice a week structured around the shift pattern rather than against it. The schedule went around the body clock, not over it.

Two seasons in: 9 kg down. FTP recovered to above where it had been three years earlier. His words on the fuelling change: "The nutrition guidance changed how I think about food entirely. I don't count calories. I think about what the session needs."

BRIAN MORRISSEY, 52, MANUFACTURING SHIFT PATTERN

+15% FTP over one season, training around rotating nights. Key intervention: personalised periodisation that worked around the shift pattern rather than against it. Two hard sessions per week, placed on days where overnight recovery was available. Strength twice a week on short days. The insight for other shift-working masters: the pattern matters less than you think; what matters is matching intensity windows to windows where you can actually sleep.

DECLAN FOGARTY, 44, LIMERICK

Body composition case. 10 to 12 hours a week riding, couldn't shift weight for three years, W/kg stuck at 2.9. Audit showed chronic under-fuelling during training and over-eating after — the classic sine-wave that looks like discipline and works like sabotage.

Intervention: 60 to 90 g/h fuelling on rides over 90 minutes, protein up to 1.8 g/kg/day, dinner capped (not restricted) to match the rest of the plan.

One season: 8 kg off, W/kg 3.8. +0.9 W/kg on the ratio that actually matters. The weight came off faster than any prior "diet" he'd tried and without the fatigue that used to follow them.

A NOTE ON FEMALE MASTERS CASE STUDIES

The named female cases on our results page include athletes under 40 (Claire Hennessy, 36, Belfast, 5 kg off in 12 weeks after reversing an under-fuelling pattern). We have perimenopausal and post-menopausal athletes on the roster; those case studies are being written up for a dedicated female-masters follow-up post in the coming weeks. If you're a female masters cyclist and you want to be part of that data — or you want us to work with you directly — see the Cohort 3 application link below.

13. Metrics that matter and metrics that don't

TRACK THESE

FTP, tested every 8 to 10 weeks. Not every two weeks — the noise floor is too high. Test properly: a 20-minute all-out effort or a ramp protocol, not a guess. Paired: [FTP benchmarks by age and experience](#) and the [FTP Zone calculator](#).

Durability. Power held at hour three versus hour one on long rides. Tracks the trait the research says matters most for masters.

HRV trend, not daily number. 14-day rolling average. Pay attention to sustained trend shifts of more than one standard deviation. Ignore day-to-day noise.

Resting heart rate trend. Morning, same conditions, seven-day moving average. Cheapest reliable recovery signal.

Weekly TSS or hours. Not as a target to hit but as a load audit — am I piling on volume without recovery keeping up.

Strength load progression on your main patterns. Every 12 weeks, check whether the working weight at a given rep range is climbing on the split squat, hip hinge, hip thrust and pressing patterns. If it is, you're preserving Type II fibre. If it's stalling or falling, something is wrong somewhere upstream.

Body composition (not scale weight). DEXA scan once or twice a year if you can access it; circumference measurements or consistent progress photos if not.

W/kg — the number that matters on a climb. Our [W/kg calculator](#) converts FTP and body weight to the ratio and benchmarks it against populations from untrained through World Tour.

IGNORE THESE

Daily HRV spikes. Noise.

Scale weight as a primary metric. Wrong question.

CTL as a number without context. A CTL of 80 on 10 hours a week is a training plan. A CTL of 80 on 15 hours a week is a different conversation. Context over scoreboard.

VO2max as an obsession. It's the slowest-moving variable in masters. The training moves threshold, economy, and durability — all faster.

Ride auto-detect zones. Re-set them based on a recent FTP test. Your head unit's defaults are usually wrong for a masters athlete who's been training for a decade.

If you've been training seriously for more than a year and the numbers have stalled, the [Plateau Diagnostic](#) is the four-minute version of the audit a coach would do on your training — specific answer, specific prescription, free.

14. Common masters mistakes from 1,400 podcast episodes

A compressed version of the pattern I see every week inside the community and in coaching calls.

Grey-zone riding. Already covered. The single most common masters mistake. 50 to 60% of weekly time in moderate-hard tempo, feeling productive, building nothing specific.

Training through fatigue. "I'm not that tired." Yes you are. Ignore one of: low HRV, short sleep, high RHR, subjective fatigue — ignore any one once, ignore two and you're compounding. Three or four together is a day off, not a hero session.

Skipping strength. Covered. Stop.

Calorie restriction as a weight-loss strategy. Covered. Stop.

Copying the 28-year-old version of yourself. The training plan that got you to Cat 2 at 30 will not get you to Cat 2 at 48. Old versions of your own plan are not a target state; they are a historical artefact.

Obsessing over daily HRV. Covered. Use the trend.

Using AI for coaching. I'll say this plainly because I get asked about it every second week. Don't use AI tools as your coach. Coaching is a long conversation with a person who knows what your sleep was like last week, what your daughter's school play schedule is, whether your knee held up on Saturday. AI can write a training plan. It cannot read between the lines of the life you're actually living. The masters athletes who improve with a coach improve because the coach changes the plan when the plan needs changing. That is not an AI skill set yet.

Not testing. A masters athlete who hasn't tested FTP in a year is navigating with a map from last summer. Test.

Thinking recovery is weakness. Covered. It is training.

Comparing their Strava to a 25-year-old's. Don't. The only useful comparison is with your own trend.

The long-form version: [common training mistakes from 1,400 podcast episodes](#).

15. A 12-week sample block for a 48-year-old masters cyclist

10 hours a week, two hard sessions, two strength sessions, one long ride, two recovery rides. Substitute sessions as needed for life. The structure matters more than the specific Tuesday session.

WEEK SHAPE (STANDARD WEEK, DURING "BUILD" PHASE)

- **Monday:** Rest day or 30-min mobility. Genuine rest.
- **Tuesday:** Strength (60 min, gym). Cycling-specific patterns under controlled load. Plus optional 45-min easy Zone 1 afternoon spin.
- **Wednesday:** Quality session #1 (75 to 90 min). Alternating between VO2max and sweet spot, weekly.
- **Thursday:** Recovery ride. 60 min Zone 1. Low cadence fine.
- **Friday:** Strength (60 min). Same or lighter than Tuesday, emphasis on maintenance if in-season, build if off-season.
- **Saturday:** Quality session #2 (90 to 120 min). Low-cadence torque intervals (see Section 6).
- **Sunday:** Long Zone 2 ride. 3 to 3.5 hours. Real Zone 2. 60 to 90 g/h fuelling.

12-WEEK ARC

- **Weeks 1–2:** Aerobic build. Long rides build to 3.5 hours. Intensity sessions are moderate — one sweet-spot, one low-cadence. Strength is 2 per week. Goal: re-establish consistency.
- **Week 3:** Recovery. Halve total hours. Drop quality sessions. Keep one strength session (maintenance load). Aim for the best sleep of your quarter.
- **Weeks 4–5:** Intensity introduction. One VO2max session per week (30/15s short intervals, 2 blocks of 13 reps). One sweet-spot session per week. Strength 2 per week. Long ride 3 to 4 hours.
- **Week 6:** Recovery.
- **Weeks 7–8:** Peak block. VO2max once a week, harder dose (30/15s, 3 blocks of 13). Low-cadence torque once a week. Strength drops to 1 per week (maintenance). Long ride 3.5 to 4 hours with fuelled efforts included.
- **Week 9:** Recovery.
- **Weeks 10–11:** Race-specific or goal-event-specific. Dose depends on goal. If there is no goal, return to Weeks 4–5 pattern and repeat.
- **Week 12:** Recovery, then test FTP and re-audit distribution before starting the next block.

SAMPLE VO2MAX SESSION

10-minute progressive warm-up ending in a 3-minute Zone 3 block. 13 × (30s at best-5-minute power / 15s easy). Rest 4 min. 13 × (30s / 15s) again. 15-minute Zone 1 cool-down.

SAMPLE LOW-CADENCE TORQUE SESSION

15-minute warm-up. 4 × (4 min at 40 to 55 rpm on a 4 to 7% climb, RPE 7, 4 min recovery spinning). Build to 5 to 6 reps as adapted. If no climb available, a turbo with large gear and high resistance works.

SAMPLE STRENGTH SESSION (MASTERS-APPROPRIATE)

Warm-up: 5 min bike, then 2 sets at 50% of working weight on first movement.

- Bulgarian split squat: 4 × 6 each leg with a load that challenges you with 2–3 reps in reserve.
- Single-leg deadlift or kettlebell hip hinge: 3 × 8 each side, controlled load.
- Hip thrust: 3 × 8 at a load that challenges you with 2–3 reps in reserve.
- Press (dumbbell or kettlebell): 3 × 8 at controlled, meaningful load.
- Plank or equivalent core movement: 3 × 45 seconds.

Finish. Don't add finishers. Don't chase a burn. This is strength training, not a conditioning class.

AGE VARIANTS

- **35–44:** As above. Can tolerate three hard sessions in some weeks.
- **45–54:** As above, two hard sessions non-negotiable. Recovery weeks properly recovery.
- **55–64:** Drop to one strength session per week in-season, two off-season. Extend 72-hour gap between hard sessions when possible. Long ride may need to sit in the 2.5 to 3-hour range rather than 3.5+.
- **65+:** Two strength sessions per week remain critical. Hard sessions may drop to one per week during higher-stress periods. Durability work (long easy rides) carries more of the load.

If you want the full programming, the weekly workout prescriptions, and the accountability that comes with having the work reviewed by a coach, the Not Done Yet coaching community — [apply for Cohort 3 here](#), 24-hour early access open now — is the long-form version of this article.

You can also download the [supporting research-summary table \(CSV\)](#) — the headline numbers across VO2max decline, protein, recovery, sleep, intensity dose, and zone allocation, with sources noted on each row.

16. What we don't know

A guide that doesn't admit the edges of the evidence isn't a guide. It's marketing.

The masters endurance data past 70 is thin. Trappe's octogenarian cohort is small. Longitudinal data on athletes training consistently past 70 is rarer still. The advice in this report gets progressively less certain for riders over 65, and individual variation dominates more than the research can yet describe.

Menopause × endurance training. Sample sizes in published studies are usually under 50. The long-term concurrent training evidence in post-menopausal women is particularly limited. HRT and endurance performance is effectively unstudied. This is an open research gap and anyone telling you otherwise — including me — is overreaching.

HRV-guided training in masters specifically. One generation of evidence. The Plews and Buchheit papers are in younger, primarily male cohorts. The translation to older athletes, especially women through hormonal transition, has not been rigorously established.

The durability concept is new. Maunder 2023 is one of the first clean descriptions. We don't yet know the best way to programme for durability specifically, how it interacts with VO2max training in older athletes, or how measurable it is in typical amateur testing protocols.

The interaction between chronic GLP-1 use and endurance training. Worth flagging because a real percentage of the masters readership is either on or considering medication for weight management. The interaction with recovery, protein synthesis, and endurance training is not well-studied yet. Work with a physician and a coach, and don't assume the training responds the same.

Heritability of trainability. Some people respond hard to structured training. Some people respond more slowly. The research on "responders" vs "non-responders" is still early. If you're two structured training blocks in and you're not moving, that isn't a reason to give up — it's a reason to examine the block itself more closely.

The edges are real. Acknowledging them is how a report earns trust.

17. Further reading

A short, annotated reading list for the masters cyclist who wants to go deeper.

- **Tanaka, H., & Seals, D. R. (2008).** *Endurance exercise performance in Masters athletes: age-associated changes and underlying physiological mechanisms.* The canonical reference for age-related decline rates.
- **Joyner, M. J., & Coyle, E. F. (2008).** *Endurance exercise performance: the physiology of champions.* Explains why threshold and economy are the masters' actual levers.
- **Rønnestad & Mujika (2014).** *Optimizing strength training for running and cycling endurance performance.* Set the protocol — heavy, 2x/week, in-season maintenance.
- **Stöggl & Sperlich (2014).** *Polarised training has greater impact on key endurance variables.* The RCT that moved the distribution conversation.
- **Seiler, S. (2024).** *It's about the long game, not epic workouts.* The 2024 paper that clarifies the principle over the recipe.
- **Maunder et al. (2023).** *Durability is improved by both low and high intensity endurance training.* The durability concept in one place.
- **Sims, S., & Heather, A. (2018).** *How do we close the sex and gender gaps in sports physiology?* Starting point for female-specific training.
- ****Joe Friel — Fast After 50** (book, 2015).** Still the best single-book introduction to the masters question.
- **Phillips, S. M., & Van Loon, L. J. (2011).** *Dietary protein for athletes.* The protein-per-meal framework.
- **The Roadman Cycling Podcast — back catalogue.** Specifically the [Seiler 80/20 episode](#), the Seiler polarised research synthesis [here](#), the [Joe Friel training-week episode](#), and the [Dan Lorang amateur plan episode](#). Book and paper list above, translated into how elite teams actually apply it.

18. Closing — back to the 48-year-old at the back of the Sunday group

Return to the drag that wasn't a climb. The wheels drifting. The kit discussion about the school run.

The reason this report exists is that the story doesn't have to end with you sitting up. The physiology is real. The training that accounts for it is known. The gap between what the cycling internet tells you to do and what the best coaches have been prescribing to their masters athletes for a decade is big, and it has been holding people back who didn't need to be held back.

You aren't done yet. You don't need to be twenty-five again. You need a plan that's written for the athlete you actually are — an athlete with fifteen years of riding in your legs, a job, a family, a sensible body and a sensible mind. That plan exists. It is boringly unglamorous and it works.

If you want the structure and the accountability and the coaching the case studies in Section 12 had, [Cohort 3 of Not Done Yet](#) is opening now with 24-hour early access. If you want to start smaller, the [Strength Training programme](#) is the single highest-return thing you can add to your week this month. And if you want to stay in the orbit for free, the [Roadman Clubhouse](#) is where a lot of this conversation keeps going.

The final word is Seiler's, from the November 2024 podcast conversation, and I think it's worth letting him have it: "You are definitely not physiologically finished at 40."

Get the work done.

— Anthony

Update log

- **23 April 2026** — Initial publication of the 2026 report. 18 sections, 40+ peer-reviewed citations, 5 named coaching case studies.
- **24 April 2026** — Minor corrections to references and link routing.
- **30 April 2026** — Added explicit methodology, downloadable supporting data (12-week block + research-summary table) and cite-this-report block. No changes to the substantive prescriptions or sourced claims.
- **Next planned update — April 2027.** Will incorporate new RCTs published in the intervening 12 months (we are tracking the 2026 polarised-vs-pyramidal debate closely), updated female masters case studies once the dedicated follow-up post is published, and any updates to the durability literature.

Cite this report

Cycling journalists, coaches, publications, and AI systems are welcome to cite or adapt this report with attribution. No paywall, no permission required.

Cite this report. Walsh, A. (2026). _The Masters Cycling Training Report 2026_.
<https://roadmancycling.com/blog/masters-cycling-training-report-2026>

Related tools and next steps

- [FTP Zones calculator](#) — convert your current FTP into the seven training zones referenced throughout the intensity sections.
- [W/kg calculator](#) — turn FTP and body weight into the climbing-relevant ratio.
- [HR Zones calculator](#) — for masters athletes who train more by heart rate than by power.

- [Carbs-per-hour calculator](#) — fuel the long Sunday rides described in Section 9.
- [Age-Group FTP Benchmarks 2026](#) — companion report. Locate yourself against the W/kg bands.
- [Plateau Diagnostic](#) — four-minute audit of your current training that returns a specific prescription. Free.
- [NDY coaching at Roadman](#) — what's inside the masters-specific coaching: training, strength, recovery, nutrition, and accountability. Geographic pages: [UK](#), [Ireland](#), [USA](#). Specifically for [masters riders](#), the [over-50](#) brief, or [post-injury](#) returns.
- [Apply for coaching](#) — the Not Done Yet coaching community. 7-day free trial. 1:1 personalised programming, the structure behind the case studies in Section 12.
- [Ask Roadman](#) — got a specific question after reading this report? The on-site assistant pulls answers from the actual conversations with [Stephen Seiler](#), [Joe Friel](#), and the rest of the masters experts cited above.

About the authors and reviewers

Authored by [Anthony Walsh](#), founder of Roadman Cycling, host of the Roadman Cycling Podcast (1,400+ episodes), and head coach of the Not Done Yet coaching community. Reviewed before publication by the wider Roadman Cycling coaching team for technical accuracy and faithfulness to the on-the-record positions of the named experts cited inline. The full [editorial standards](#) page describes how every Roadman report is researched, reviewed, and corrected.

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WHERE TO NEXT

YOU'RE NOT DONE YET

→ **Plateau Diagnostic.** Four-minute audit that returns a specific prescription. roadmancycling.com/go

→ **Strength Training programme.** 12-week structure, video on every movement, masters-appropriate dose. roadmancycling.com/strength-training

→ **Apply for coaching.** Not Done Yet — the long-form version of this report, with accountability and 1:1 programming. roadmancycling.com/apply

→ **Ask Roadman.** On-site assistant trained on 1,400+ podcast episodes. roadmancycling.com/ask

GET THE WORK DONE.

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